

HARIHARAN T S

Full Stack Engineer (AI & Cloud) | FastAPI · React · LLM Integration · Cloud Deployment

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PROFILE SUMMARY

Recently graduated Full Stack Engineer with hands-on experience building and deploying production-grade, AI-integrated web applications. Developed a cloud-backed full-stack system using Python (FastAPI) and React, integrating ML models, LLM-powered features, RAG pipelines, and prompt engineering — achieving 91% model accuracy with sub-200ms latency across 10,000+ records. Experienced in REST API design, SQL/NoSQL databases, cloud deployment (Firebase, Vercel), and CI/CD via GitHub. Familiar with Node.js, TypeScript, Docker, JWT-based authentication, and AWS fundamentals. Actively uses AI-assisted development tools (GitHub Copilot, Cursor) and brings a strong product-focused, ownership-driven mindset to every build.

TECHNICAL SKILLS

- ▶ **Backend:** Python (FastAPI) — Primary · Node.js (Familiar) · NestJS (Familiar) · REST API Design · Async I/O · Microservices Architecture · Request Validation · Modular Pipeline Architecture
- ▶ **Frontend:** React.js · JavaScript · TypeScript (Familiar) · Context API · Canvas API · Component-based UI · Dynamic Rendering · Responsive Design
- ▶ **AI & LLM Integration:** TensorFlow · Scikit-learn · NLP · OCR (Tesseract) · CNNs · Prompt Engineering · RAG Pipelines · LLM Integration · Vector Embeddings (FAISS/ChromaDB) · OpenAI API (Familiar) · LangChain (Familiar) · ML Inference & Evaluation
- ▶ **Databases:** MySQL · SQLite · Firebase (NoSQL/Realtime DB) · PostgreSQL (Familiar) · MongoDB (Familiar) · SQL Query Optimisation · Offline-first Sync Architecture
- ▶ **Cloud & DevOps:** Firebase (Cloud) · Vercel (Deployment) · AWS Fundamentals (EC2, S3, Lambda — Familiar) · Docker (Basic) · GitHub Actions (CI/CD) · Git · GitHub
- ▶ **Security:** JWT Authentication (Familiar) · OAuth2 (Familiar) · API Security Best Practices via FastAPI
- ▶ **AI Dev Tools:** GitHub Copilot · Cursor AI · Bolt (Familiar) · AI-assisted Debugging & Prototyping
- ▶ **Languages:** Python · JavaScript · TypeScript · Java · SQL
- ▶ **Core CS:** Data Structures & Algorithms (400+ LeetCode) · OOP · System Design · SDLC

PROJECTS

AI-Powered Personal Finance Management System → Live, Cloud-Deployed App

Jan 2026 – Present

Stack: Python · FastAPI · TensorFlow · Scikit-learn · Tesseract OCR · NLP · SQLite · Firebase · React · Prompt Engineering · RAG · Vector Embeddings

- ▶ **Architected and deployed a production-grade full-stack AI application** — designed the complete pipeline (SMS/receipt ingestion → OCR → NLP → ML classification → FastAPI microservice → React dashboard) following a decoupled microservices pattern, deployed on Firebase (cloud backend) and Vercel (frontend).
- ▶ **Built 12+ async FastAPI (Python) REST endpoints** with modular architecture, API contract documentation, request validation, and async I/O handling — supporting 10,000+ records with offline-first SQLite + Firebase cloud sync and fault-tolerant failover.
- ▶ **Implemented LLM integration and RAG pipeline** using vector embeddings to ground LLM responses in user-specific financial data, enabling contextual natural language Q&A — reducing hallucinations and improving response relevance; applied prompt engineering for consistent, structured GenAI outputs.
- ▶ **Trained and tuned a multi-class ML classification model** on 5,000+ transactions achieving ~91% accuracy with sub-200ms inference latency using TensorFlow and Scikit-learn; iteratively improved using data augmentation, threshold tuning, and F1-score evaluation.
- ▶ **Built a full-stack React frontend** with real-time data rendering, expense analytics dashboards, and an LLM-powered budget insight interface — focused on performance and responsive user experience.

- ▶ **Applied JWT-based API security patterns and environment-based config management** to protect sensitive financial data and follow secure development best practices.
- ▶ **Maintained CI/CD workflow via GitHub** with version-controlled pipeline stages, enabling safe iterative deployment without breaking existing functionality.

Interactive Data Explorer — One Piece Encyclopedia → Live on Vercel

May 2026 – Present

Stack: *React.js · JavaScript · Context API · Canvas API · JSON · Lazy Loading · Vercel · Git*

- ▶ **Designed and deployed a scalable React application** representing 200+ entities and 30+ story arcs using normalised relational data models optimised for efficient cross-entity querying and access.
- ▶ **Implemented global state management** with Context API across 15+ routes and built Canvas API visualizations for interactive relationship mapping — focused on clean, component-driven architecture and reusability.
- ▶ **Optimised frontend performance by ~40%** using lazy loading and memoization; delivered a smooth, demo-ready experience deployed via Vercel with CI integration through GitHub.

Traffic Sign Detection — CNN Classifier

Dec 2024

Stack: *Python · TensorFlow · CNN · OpenCV · Scikit-learn · GTSRB Dataset · Data Augmentation*

- ▶ **Developed an AI/ML model** — CNN trained on 50,000+ images across 43 classes achieving 96.4% test accuracy; applied data augmentation and class-weighted loss functions, reducing misclassification by 22%.
- ▶ **Evaluated model performance** using precision, recall, F1-score, and confusion matrix analysis; documented benchmarks and failure-mode analysis to assess production deployment readiness.

E D U C A T I O N

B.E. Computer Science & Engineering (AI & ML) — Sethu Institute of Technology, Kariapatti

2022 – 2026

CGPA: 8.79 / 10 | Honors: Full Stack Development

Relevant Coursework: Machine Learning, Deep Learning, NLP, Data Structures & Algorithms, DBMS, Computer Vision, Software Engineering

Class XII — Thiagarajar Model Higher Secondary School, Madurai

2022 | 84%

Class X — Thiagarajar Model Higher Secondary School, Madurai

2020 | 83.4%

C E R T I F I C A T I O N S

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- ▶ AI Master Class (30-day Hands-on Program) — Novitech Pvt Ltd | Apr 2024
 - ▶ AI Primer Certification — Infosys Springboard
 - ▶ Project-Based Robotics Training (2 programmes) — Zetspire Technologies Pvt Ltd | 2023 & 2024

A C H I E V E M E N T S

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- ▶ **Top 5 Finish** — Hackathon at Mohammed Sathak Engineering College, Kilakarai.
 - ▶ **400+ LeetCode problems** solved across arrays, trees, graphs, and dynamic programming.